

Digital Network Modernization & Customer Experience Overhaul for a Telecom Operator

Consulting Case Study | Telecom Industry

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Executive Summary (TLDR)

A national telecom operator was losing postpaid subscribers to a smaller, digitally native competitor despite having superior network coverage. Legacy OSS/BSS systems, slow provisioning times, and a fragmented customer experience across app, call center, and retail were driving churn. A structured network-and-CX transformation program cut new-service provisioning time from **5 days to 4 hours**, reduced churn by **1.8 percentage points**, and lifted digital self-service adoption from **31% to 58%** within 15 months.

Table of Contents

- Background
- Methodology
- Key Considerations
- Sample Deliverables
- Strategic Recommendations
- Culture and Change Management
- Technology and OSS/BSS Modernization
- Operational Efficiency
- Regulatory Considerations
- Risk Identification
- Cost-Benefit Analysis
- Monitoring and KPIs
- Integration Roadmap
- Return on Investment
- Establishing Governance
- Implementation Playbook
- Additional Considerations
- Key Findings and FAQs

1. Background

A national telecom operator with strong network coverage and brand recognition began losing postpaid subscribers at an accelerating rate — down 1.8 points in market share over six quarters — despite consistently outperforming competitors in network quality surveys. Customer research revealed the issue wasn't the network; it was everything around it. Activating a new line took up to five business days due to manual provisioning steps across disconnected legacy OSS (Operations Support Systems) and BSS (Business Support Systems). Customers switching plans often experienced billing errors for one to two cycles. The mobile app covered only basic functions, pushing most service requests to call centers with average wait times exceeding 12 minutes.

A smaller, digitally native competitor with weaker network coverage was winning switchers specifically because activation, plan changes, and support were instant and self-service. Leadership needed to determine whether to prioritize a full OSS/BSS replacement — expensive and high-risk — or a more targeted modernization of the systems and journeys causing the most customer pain.

2. Methodology

A **6-phase approach** combining operational and digital transformation methodology was applied:

- **Phase 1 – Diagnostic & Journey Mapping:** Quantify friction points across activation, plan changes, billing, and support using both system data and customer interviews.
- **Phase 2 – Prioritization:** Rank pain points by combined customer impact and technical feasibility, rather than pursuing a full system replacement.
- **Phase 3 – Target Architecture Design:** Design a modernized provisioning layer that could sit atop legacy OSS/BSS via APIs, avoiding a full "rip and replace."
- **Phase 4 – Journey Redesign:** Rebuild the top five customer journeys (activation, plan change, billing dispute, device upgrade, outage reporting) for digital-first resolution.
- **Phase 5 – Phased Rollout:** Deploy market-by-market to manage risk and capture learnings before national rollout.
- **Phase 6 – Scale & Optimize:** Expand successful patterns nationally and retire redundant manual processes.

3. Key Considerations

- A full legacy system replacement was rejected as too risky and slow given competitive pressure; an API-layer approach delivered most of the customer-facing benefit at a fraction of the cost and time.
- Provisioning delays were traced to seven manual handoff points between systems that had accumulated over a decade of incremental patches — the fix required both technology and process redesign.
- Call-center staff needed a clear transition plan, since reduced call volume from improved self-service directly affected their role.

4. Sample Deliverables

- Customer journey friction diagnostic and prioritization matrix
- API-layer target architecture blueprint (legacy-compatible)

- Redesigned digital journeys for activation, plan change, billing, and support
- Phased market rollout plan with success criteria per wave
- Call-center transition and reskilling plan

5. Strategic Recommendations

The recommended path avoided a full OSS/BSS replacement in favor of an **API orchestration layer** that automated the manual handoffs causing provisioning delays, paired with a redesigned mobile app covering the top customer journeys end-to-end. This combination addressed 80% of customer pain points within a fraction of the time and cost of full system replacement, while preserving the option for deeper modernization later.

6. Culture and Change Management

Call-center leadership initially resisted the shift toward self-service, viewing reduced call volume as a threat to their teams. The program reframed the transition around redeploying staff toward complex, high-value support cases — the ones self-service couldn't resolve — with a structured reskilling plan rather than headcount reduction, which secured buy-in from frontline leadership.

7. Technology and OSS/BSS Modernization

Rather than replacing legacy OSS/BSS systems outright, an orchestration layer was built to automate the seven manual handoff points in the provisioning process, cutting activation time from five days to under four hours while legacy systems remained the system of record underneath.

8. Operational Efficiency

Automating manual provisioning steps eliminated a significant volume of internal rework caused by human error in the handoff process, which had previously been a major driver of billing disputes in the first two cycles after any account change.

9. Regulatory Considerations

Billing accuracy improvements were tracked against regulatory requirements for consumer billing transparency, with the redesigned billing dispute journey built to meet documented resolution-timeline requirements consistently, which had previously been missed during peak volume periods.

10. Risk Identification

Key risks included the possibility of new customer-facing bugs during phased rollout, the risk of over-automating and removing necessary human judgment in edge cases, and the risk of call-center attrition if the transition plan wasn't communicated clearly. Each was mitigated through phased deployment, defined escalation paths, and early, transparent staff communication.

11. Cost-Benefit Analysis

The API orchestration approach cost roughly a third of a full OSS/BSS replacement while delivering the majority of customer-facing improvement. The business case showed payback within 18 months driven by reduced churn and lower call-center cost-to-serve.

12. Monitoring and KPIs

Core KPIs tracked weekly included activation time, digital self-service adoption rate, first-cycle billing accuracy, call-center volume, and churn rate — allowing the team to validate impact market by market before national rollout.

13. Integration Roadmap

Rollout proceeded in three waves — two pilot markets, a mid-size regional expansion, and finally national deployment — with each wave's KPIs reviewed before proceeding, allowing course correction on journey design based on real usage data.

14. Return on Investment

Within 15 months, provisioning time dropped from five days to four hours, churn improved by 1.8 percentage points, and digital self-service adoption rose from 31% to 58%, with call-center cost-to-serve down significantly as a result.

15. Establishing Governance

A joint IT-Operations-Customer Experience steering committee was established to own the transformation, ensuring technology, process, and frontline considerations were represented in every major rollout decision.

16. Implementation Playbook

A standardized market-rollout playbook was created covering technical deployment checklists, staff training sequencing, and customer communication timing, reducing variability in how each new market wave was executed.

17. Additional Considerations

The orchestration-layer approach was explicitly designed to be a stepping stone — architected so that a future full OSS/BSS modernization, if pursued, could build on the same API layer rather than requiring another disruptive redesign.

18. Key Findings and FAQs

Why not replace the legacy systems outright? The risk, cost, and timeline of a full replacement didn't match the urgency of the competitive threat; the orchestration-layer approach delivered comparable customer impact far faster.

What was the single biggest driver of churn reduction? Faster activation and plan-change times, which directly addressed the primary reason customers cited for switching to the digital-native competitor.

Is this approach reusable for other operators with legacy systems? Yes — the API orchestration pattern is broadly applicable wherever legacy system replacement is too risky to attempt in the required timeframe.

